

COUNTEREXAMPLES TO CONJECTURE 6.18 IN SULLIVAN’S CACCETTA–HÄGGKVIST SURVEY

ABSTRACT. Sullivan recorded a conjecture attributed to Charbit asserting that every n -vertex digraph D with adjacency spectral radius $\rho(D) \geq n/k$ contains a cycle of length at most k , where, as throughout that survey, a cycle in a digraph means a directed cycle. We show that the assertion holds exactly for $k = 1, 2$, vacuously for $k = 1$. For every $k \geq 3$, the lexicographic product $\vec{C}_{k+1}[T_5]$ is a strongly connected oriented graph of directed girth $k + 1$ and spectral radius $1/(2^{1/5} - 1) > 5(k + 1)/k$, and hence is a counterexample.

Throughout, digraphs are finite, nonempty, and loopless, with no parallel arcs; antiparallel pairs are permitted. For a digraph D , let $A(D)$ be its adjacency matrix, with $A(D)_{uv} = 1$ when $u \rightarrow v$, and let $\rho(D) = \rho(A(D))$. The directed girth of D is the length of a shortest directed cycle.

Sullivan [1, Conjecture 6.18] recorded the following conjecture, attributed to Charbit: if D has n vertices and $\rho(D) \geq n/k$, then D contains a directed cycle of length at most k . For a positive integer k , denote this assertion by (C_k) .

The source states the conclusion with the bare word “cycle”: “If the spectral radius of A is at least n/k , then G has a cycle of length $\leq k$ ” [1, Conjecture 6.18]. Here, as throughout [1], a cycle in a digraph means a directed cycle. That is the convention of the survey’s headline statement, the Caccetta–Häggkvist conjecture, recorded there as “Every simple n -vertex digraph with minimum out-degree at least r has a cycle with length at most $\lceil n/r \rceil$ ” [1, Conjecture 2.1]. The survey then fixes the meaning of that conclusion itself, by restating the same conjecture in matrix form immediately afterwards: for a 0-1 matrix A in which $a_{ij} = 1$ implies $a_{ji} \neq 1$ for $i \neq j$, and $a_{ii} = 1$ for all i , “if the sum of every row in A is at least $r + 1$, $A^{\lceil n/r \rceil}$ has trace greater than n ” [1, §2]. Powers of such a matrix record directed walks and nothing else, so, the diagonal being all ones, $\text{tr } A^t > n$ holds exactly when some vertex carries a closed directed walk of length t other than the all-loops one, that is, exactly when the digraph has a directed cycle of length at most t ; a cycle of the underlying graph contributes to no power of A . The same dictionary governs the subsection in which Conjecture 6.18 appears, where a trace condition of this kind is spelled out as asking for a “non-trivial (i.e. not all horizontal edges) path” from one copy of a vertex to a later copy of the same vertex [1, Conjecture 6.17]. This reading is load-bearing rather

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than cosmetic, and Remark 3 records what the counterexamples below do and do not refute under it.

Theorem 1. *The assertion (C_k) holds if and only if $k \in \{1, 2\}$. For every $k \geq 3$, it fails even among strongly connected oriented graphs.*

1. THE CONSTRUCTION

Let $\vec{C}_g$ be the directed cycle on $\mathbb{Z}_g$, and let T_m be the transitive tournament on $[m] = \{1, \dots, m\}$, with $j \rightarrow \ell$ when $j < \ell$. Define $D_{g,m} = \vec{C}_g[T_m]$ on $\mathbb{Z}_g \times [m]$ by

$$(i, j) \rightarrow (i, \ell) \quad (j < \ell), \quad (i, j) \rightarrow (i+1, \ell) \quad (\ell \in [m]),$$

where the first coordinate is taken modulo g , and there are no other arcs.

Proposition 2. *For all $g \geq 2$ and $m \geq 1$, the digraph $D_{g,m}$ is strongly connected, has order gm , directed girth g , and*

$$\rho(D_{g,m}) = \frac{1}{2^{1/m} - 1}.$$

It is oriented whenever $g \geq 3$.

Proof. Every vertex in one block has an arc to every vertex in the next block. Following such arcs, and making a full turn when necessary, gives a directed path between any ordered pair of vertices. Thus $D_{g,m}$ is strongly connected. If $g \geq 3$, neither the arcs inside a block nor the arcs between consecutive blocks form an antiparallel pair, so $D_{g,m}$ is oriented.

Along each arc, the first coordinate either stays fixed or advances by one. Hence a directed cycle uses a positive multiple of g advancing arcs: zero is impossible because every block induces the acyclic tournament T_m . Its length is therefore at least g . Choosing one vertex from each block gives a directed g -cycle, so the directed girth is g .

Set

$$r = 2^{1/m}, \quad \lambda = \frac{1}{r-1}, \quad y_j = r^{m-j} \quad (j \in [m]),$$

and define $x_{(i,j)} = y_j$. Since $r^m = 2$,

$$\sum_{\ell=1}^m y_\ell = \frac{r^m - 1}{r - 1} = \lambda.$$

For every (i, j) ,

$$\begin{aligned} (A(D_{g,m})x)_{(i,j)} &= \sum_{\ell=1}^m y_\ell + \sum_{\ell=j+1}^m y_\ell \\ &= \lambda + \frac{r^{m-j} - 1}{r - 1} = \lambda y_j. \end{aligned}$$

Thus $x > 0$ is an eigenvector with eigenvalue λ . By Perron–Frobenius, $\rho(D_{g,m}) = \lambda$. $\square$

2. PROOF OF THE CLASSIFICATION

Proof of Theorem 1. For $k = 1$, every row sum of $A(D)$ is at most $n - 1$, and hence $\rho(D) \leq n - 1 < n$. Thus the hypothesis of (C_1) is impossible.

For $k = 2$, suppose that D has no directed cycle of length at most 2. Then D has no antiparallel pair. Let $x \neq 0$ be a nonnegative Perron eigenvector of $A(D)$. Since every unordered pair supports at most one arc,

$$\begin{aligned} \rho(D)\|x\|_2^2 &= \sum_{u \rightarrow v} x_u x_v \\ &\leq \frac{1}{2} \sum_{u \rightarrow v} (x_u^2 + x_v^2) \\ &= \frac{1}{2} \sum_{v \in V(D)} (d^+(v) + d^-(v)) x_v^2 \\ &\leq \frac{n-1}{2} \|x\|_2^2. \end{aligned}$$

Therefore $\rho(D) \leq (n-1)/2 < n/2$, proving (C_2) .

Now let $k \geq 3$ and take $D = D_{k+1,5}$. By Proposition 2, D is a strongly connected oriented graph of order $n = 5(k+1)$ and directed girth $k+1$. Moreover,

$$\rho(D) = \frac{1}{2^{1/5} - 1} > \frac{20}{3} \geq \frac{5(k+1)}{k} = \frac{n}{k}.$$

Indeed, $23^5 > 2 \cdot 20^5$ gives $2^{1/5} < 23/20$ and hence the strict inequality, while

$$\frac{20}{3} - \frac{5(k+1)}{k} = \frac{5(k-3)}{3k} \geq 0.$$

Thus D satisfies the spectral hypothesis but has no directed cycle of length at most k . Hence (C_k) fails for every $k \geq 3$. $\square$

Remark 3 (Reading and scope). What is refuted is (C_k) for $k \geq 3$, on the directed reading of “cycle” fixed above, and only that. The refutation does not survive replacing directed cycles by cycles of the underlying graph: each block of $D_{k+1,5}$ induces an underlying K_5 , so the underlying graph of $D_{k+1,5}$ contains triangles, and $3 \leq k$ for every $k \geq 3$.

The counterexamples are, on the other hand, insensitive to the other matrix convention that appears in [1], the unit diagonal used in the restatement of the Caccetta–Häggkvist conjecture there. Conjecture 6.18 fixes a zero diagonal, which is the convention used above; but for the positive eigenvector x of Proposition 2 one has $(A(D) + I)x = (\rho(D) + 1)x$, so $\rho(A(D) + I) = \rho(D) + 1 > \rho(D) \geq n/k$ for $D = D_{k+1,5}$ and $k \geq 3$. The spectral hypothesis therefore holds under either normalisation, the diagonal entries being a feature of the matrix and not arcs of D , which remains loopless of directed girth $k+1$.

No minimality is claimed for $D_{k+1,5}$, either within this family or among digraphs generally, and nothing here bears on the Caccetta–Häggkvist conjecture [1, Conjecture 2.1] itself, whose minimum-outdegree hypothesis the family fails, as recorded below.

Remark 4. For $k = 5$, the smaller graph $D_{6,2}$ has order 12, directed girth 6, and spectral radius $1 + \sqrt{2} > 12/5$. The construction does not satisfy the Caccetta–Häggkvist minimum-outdegree hypothesis: for every m ,

$$\delta^+(D_{k+1,m}) = m < \frac{m(k+1)}{k} = \frac{|V(D_{k+1,m})|}{k}.$$

REFERENCES

- [1] B. D. Sullivan, *A summary of problems and results related to the Caccetta–Häggkvist conjecture*, AIM Preprint 2006-13 (2006), arXiv:math/0605646; also circulated by the American Institute of Mathematics, with the title words in the order *results and problems*, at aimath.org/WWN/caccetta/caccetta.pdf, where the item cited here is printed as “Conjecture 6.18. (Charbit)”.